

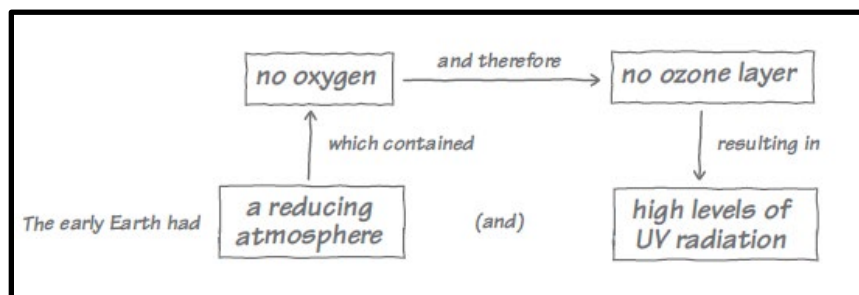
THE HISTORY OF LIFE ON EARTH

Activity 25.1 What do we know about the origin of life on Earth?

Construct a concept map of conditions on the early Earth and the origin of life forms. Be sure to include definitions or descriptions of all the terms in the list below. Keep in mind that there are many ways to construct a concept map.

- Begin by writing each term on a separate sticky note or piece of paper.
- Then organize the terms into a map that indicates how the terms are associated or related.
- Draw lines between terms and add action phrases to the lines that indicate how the terms are related.

Here is an example:



no oxygen

sunlight

ozone layer

DNA

carbon dioxide

high-oxygen atmosphere

amino acids

nucleic acids

organic compounds

prokaryotes

energy source

anaerobic bacteria

antibiotics

reducing atmosphere

electrical discharge (i.e., lightening)

sugars

inorganic compounds

no oxygen

electrical discharge (i.e., lightening)

ozone layer

DNA

carbon dioxide

RNA world

carbon source

cyanobacteria (blue-green algae)

penicillin

high-oxygen atmosphere

amino acids

nucleic acids

organic compounds

reducing atmosphere

sunlight

sugars

inorganic compounds

low levels of UV radiation

Eukarya

mode of nutrition

Gram stain

What type of concept map can you generate using some of the terms above? Sketch below:

Use the understanding you gained from creating the concept map above to answer the questions below.

1. Modern theory suggests that the early (pre-life) atmosphere on Earth was a reducing one. Why (for what reasons) is it believed that oxygen was not present when life formed on Earth?

One key piece of evidence is the lack of oxidized iron in rocks that are more than 2.7 billion years old. Therefore, oxygenic photosynthesis is thought to have evolved about 2.7 billion years ago. As oxygen accumulated in the oceans, iron oxides were formed and precipitated out. Eventually enough oxygen was formed not only to saturate the ocean water but also to outgas or move from the ocean into the atmosphere. As oxygen levels rose in the atmosphere, iron compounds on land also oxidized. (Note: Rust is a form of oxidized iron.)

2. What proposed energy sources existed on this early (pre-life) Earth?

Energy sources on the early Earth included geothermal energy, volcanic energy, and lightning.

3. In the 1950s, Stanley Miller performed a set of experiments to determine whether life could have evolved given the conditions stated in the answers to questions 1 and 2.

- a. How was the experiment designed?

Miller created a sealed system. The only compounds in the system were water and a number of the different atmospheric gases thought to have been present on the early Earth—for example, methane, ammonia, and hydrogen. Miller used heat to vaporize the water. The steam passed through the chamber that contained the atmospheric gases. The vapor cooled, and the water was collected and recirculated to be heated again to produce steam.

- b. What were the necessary controls?

The system had to be meticulously cleaned and sterilized to be sure it contained no organic molecules. This included evacuating all the air out of the system and replacing it with the ammonia, methane, and hydrogen. Another control was an identical system that was provided with no energy source. This would allow Miller to determine whether the macromolecules produced were contaminants or direct results of the experiment. In subsequent experiments, Miller also varied the concentrations and presence or absence of various compounds to determine which were required to produce specific types of macromolecules.

- c. What was produced in the experiment?

In the first experiment, organic molecules, including some amino acids, were found in the system after only about 10 days of operation. In subsequent experiments, Miller was able to produce additional amino acids, carbohydrates, nucleotide bases, and other macromolecules by adjusting the gas composition of the atmosphere in the system.

d. What did the results imply about the possible origin of life on Earth?

Miller's experiments showed that organic compounds could be produced in an anaerobic, reducing atmosphere. The results implied that life could have evolved under these conditions as well.

e. There is general agreement that life must have evolved in the oceans originally and only much later invaded land. What factors of the physical environment on the early and evolving Earth support these ideas? Changes in which of these were essential to allow life to survive on land?

The early Earth's atmosphere contained little or no oxygen. No ozone layer existed in the upper atmosphere. Though many of the early organisms on Earth did not require oxygen, with no ozone layer, the Earth was bombarded by high levels of UV light, which can easily mutate DNA and therefore kill organisms. Some scientists estimate that life on land was not possible until the atmosphere reached a minimum of about 10% oxygen and the ozone layer was formed.

f. Most of us can't imagine a world without oxygen. However, as you learned earlier, chemically oxygen is a powerful oxidizing compound. What effect(s) would the increase in oxygen levels of the atmosphere have on the organisms that existed at that time?

Most anaerobic organisms (e.g., prokaryotes) alive at the time would be unlikely to have mechanisms to counteract the oxidizing effects of oxygen. As a result, most of them would have died off. Only those that lived in anaerobic environments that were not exposed to oxygen and those that had random mutations that equipped them with antioxidant capabilities (e.g., peroxidases) would have survived.

Activity 25.2 How do we determine the age of fossils and rocks?

To determine the age of fossils and rocks, scientists determine the amounts of radioactive compounds and their stable daughter products present in the sample. Radioactive elements are known to decay into stable daughter compounds at specific rates. A number of radioactive compounds, their stable daughter compounds, and their half-lives are shown in the table below.

Radioactive Compound	Stable daughter compound	Half-life
Carbon 14	Nitrogen 14	5370 years
Potassium 40	Argon 40	1.25 billion years
Rubidium 87	Strontium 87	48.8 billion years
Thorium 232	Lead 208	14 billion years
Uranium 235	Lead 207	704 million years
Uranium 238	Lead 206	4.47 billion years

For dating **rocks**, the potassium-argon method is often used because:

- Argon is a gas. When rock is molten, any existing argon gas will escape. As a result, newly formed rocks contain no argon.
- As the ^{40}K naturally occurring in the rock decays to ^{40}Ar , the ^{40}Ar is trapped in pocket in the rock. The ratio of ^{40}K to ^{40}Ar can be measured to determine the rock's age.

For dating **organic material**, carbon-nitrogen dating is often used because:

- The ratio of radioactive to nonradioactive carbon dioxide in the atmosphere is fairly constant over time. As a result, the levels of these elements in organic tissue remain relatively constant as long as the organism is alive.
- Once the organism is dead, no new inputs of carbon occur and the existing radioactive-carbon-to-nonradioactive-carbon ratio will decrease over time as the radioactive carbon decays.

1. In one half-life, half of the original radioactive compound will decay into its stable daughter compound.

- a. If a newly formed rock contains 100 units of ^{40}K , how many units of ^{40}K would it contain after 1.25 billion years?

It would contain 50 units.

100 units ^{40}K >>>> 1.25 BY (one $\frac{1}{2}$ life) >>>> 50 units ^{40}K

- b. How many units of ^{40}Ar would the rock contain when newly formed versus after 1.25 billion years?

The rock would contain 0 units of ^{40}Ar when newly formed and 50 units after 1.25 billion years.

Why so? ^{40}Ar is NOT present until the decay process takes place. After one $\frac{1}{2}$ life, one-half of the original 100 units would be present or 50 units ^{40}Ar after 1.25 billion years.

c. After the 1.25 billion years, what would be the ratio of ^{40}K to ^{40}Ar in the rock?

Half of the ^{40}K would be gone. The ratio would be 50 units of ^{40}K to 50 units of ^{40}Ar , or 50:50 or 1:1.

d. After 2.5 billion years, what would be the ratio of ^{40}K to ^{40}Ar ?

After the next 1.25 billion years, 25 of the 50 remaining ^{40}K molecules would decay to ^{40}Ar . In other words, only 1/4; of the original ^{40}K would be present and the ratio would now be 25 units of ^{40}K to 75 units of ^{40}Ar or 1:3.

2. You are fossil hunting and find a trilobite fossil in an old riverbed. You have it radiometrically dated and are told it is 275 million years old.

a. In this amount of time, how many half-lives of ^{40}K have elapsed?

$275 \text{ million} / 1.25 \text{ billion} = 2.75 \times 10^8 / 1.25 \times 10^9 = \text{about } 2.2 \times 10^{-1} \text{ or } 0.22 \text{ half-lives.}$

b. Given your answer in part a, what would be the ratio of ^{40}K to ^{40}Ar found in the fossil remains?

In 0.2 half-lives, 0.2 of a half-life has elapsed. Therefore, 0.2 of half of the ^{40}K would have decayed to ^{40}Ar , or 0.1 or 1/10 would have decayed to ^{40}Ar . As a result, the ratio of ^{40}K to ^{40}Ar would be 9 to 1. $0.2 \times \frac{1}{2} = 0.1$ or 1/10 decayed to ^{40}Ar . Therefore, 9/10 would be ^{40}K , a ratio of 9:1 of ^{40}K : ^{40}Ar

3. You want to date some fabric that you have discovered at an archeologic dig.

a. What method of dating would be best for this? Describe the general procedure for dating the cloth.

Since human civilizations arose fairly recently in geologic time, it would be best to use carbon dating for the cloth sample (which is ORGANIC). The lab would need to test the relative proportions of carbon 14 to carbon 12 in the sample. Carbon 14:Carbon 12 ratios in organic material will decrease over time proportional to the half-life of carbon 14.

b. Assume the cloth is 2,000 years old. How would the level of the radioisotope you used to date it have changed to date it have in this period of time?

Two thousand years is $2,000/5,370$ or about 0.372 of the half-life for carbon 14. As a result, 0.372×0.5 , or about 0.186 or 19%, of the normal level of carbon 14 would have been removed, leaving the carbon 14 level at about 81% of normal.